AAYUSHI GUPTA(2301010189)

Assignment 2: Packet Flow Visualization Using Simulation Mode

Experiment Objectives:-

Students will use Packet Tracer Simulation Mode to observe how packets move between devices,how switches learn MAC addresses, and how ICMP echo request/reply travels through a small network.

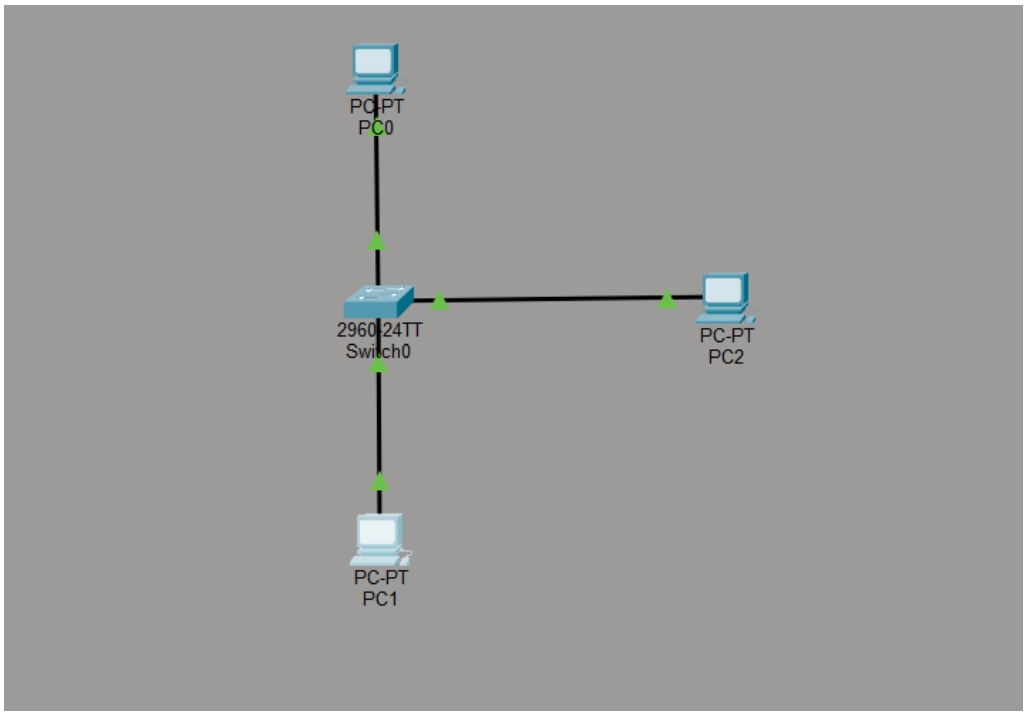
Learning Outcomes:-

- Use Simulation Mode effectively
- Observe ARP + ICMP behavior
- Understand switch MAC learning behavior
- Interpret packet event list timeline

Concepts Used:-

- Simulation Mode
- ICMP ping
- ARP resolution
- Switch MAC address table learning

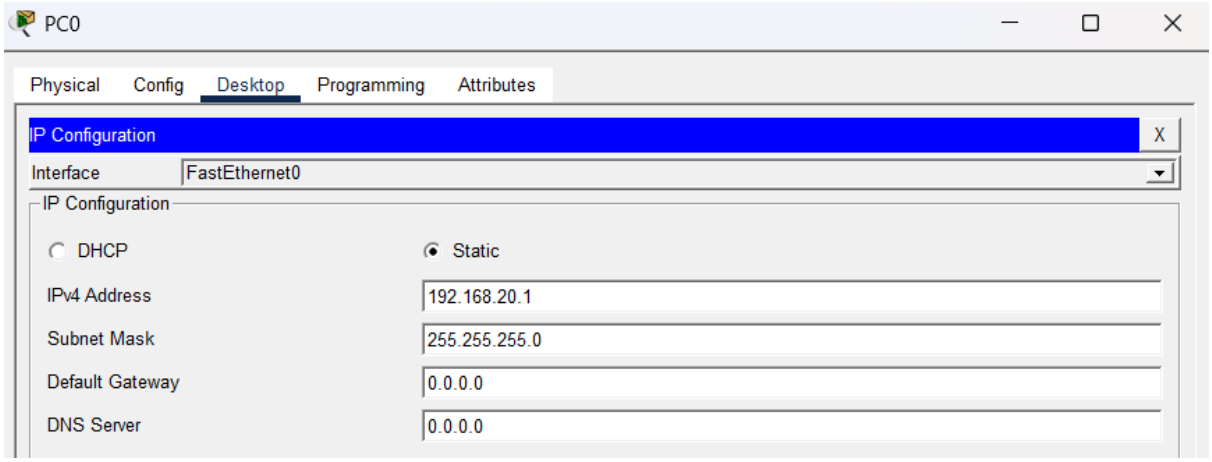
Task 1: Build a Simple LAN

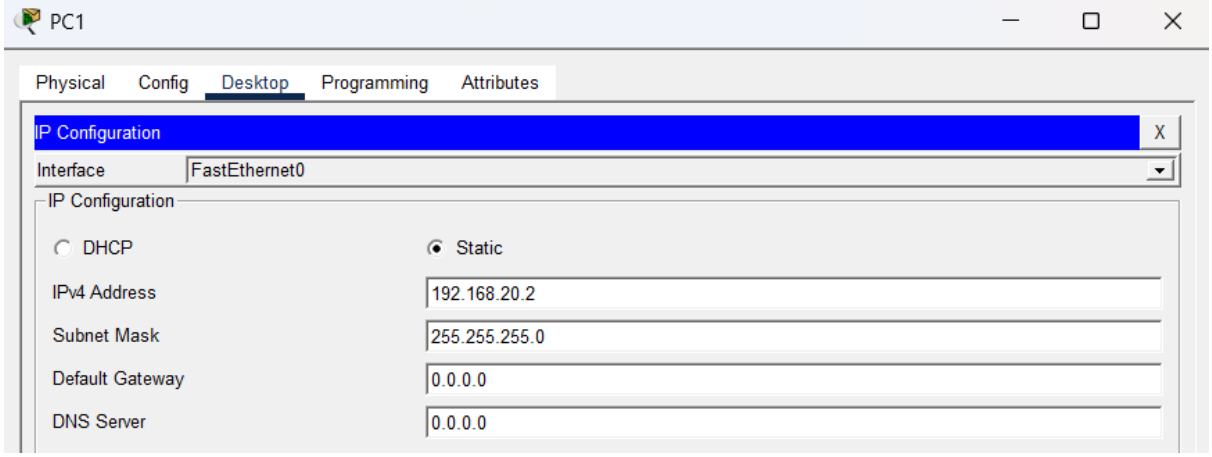


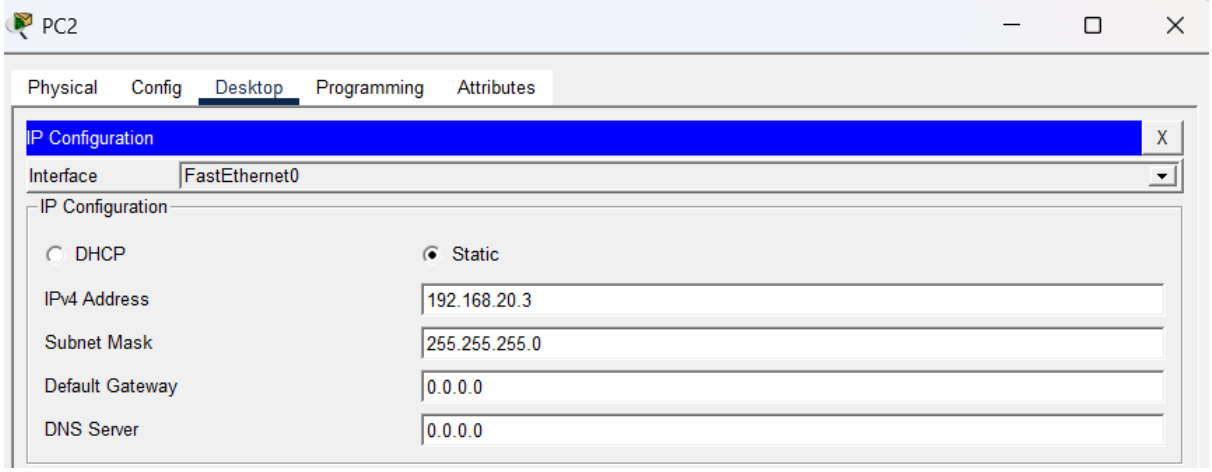
Configuration:-

- 1 Switch (2960)
- 3 PCs connected to switch
- IP addresses assigned:

PC	IP Address	Subnet Mask
PC0	192.168.20.1	255.255.255.0
PC1	192.168.20.2	255.255.255.0
PC2	192.168.20.3	255.255.255.0





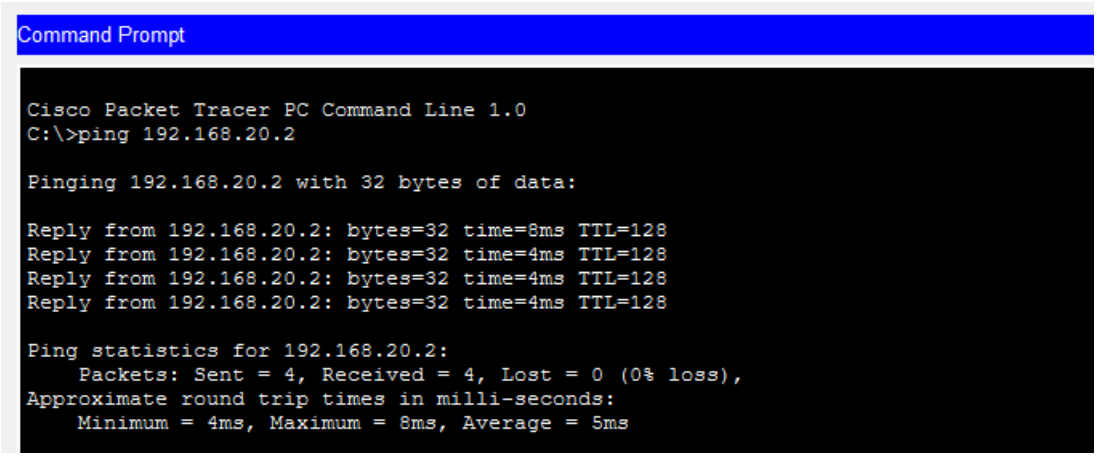


Task 2: First Ping Observation

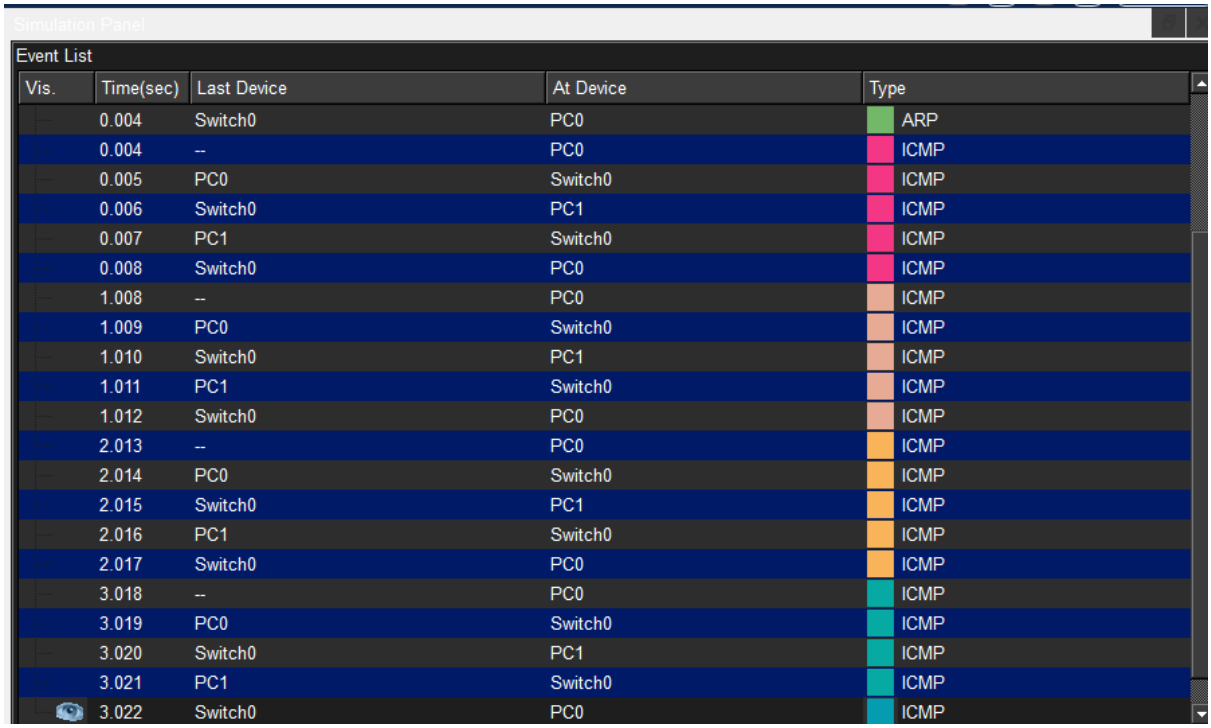
- Simulation Mode was enabled

- The Event List filter was set to show only ARP and ICMP

- A ping was issued from PC1 to PC2 via the Command Prompt: ping 192.168.20.2

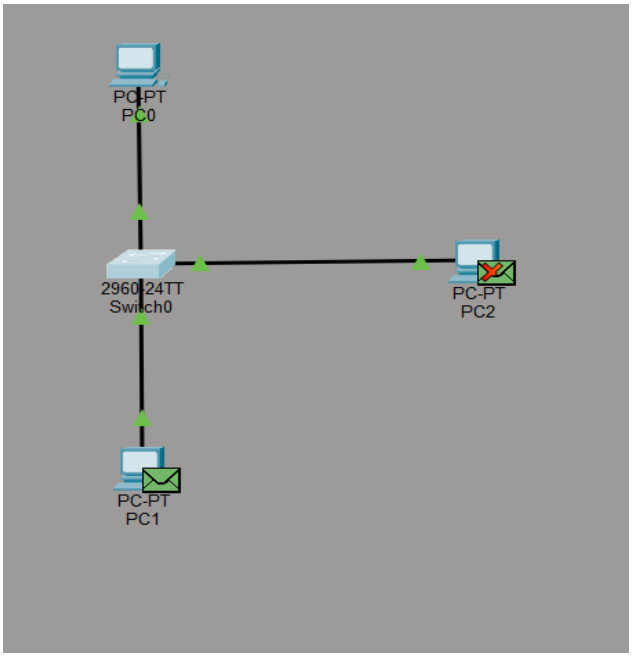


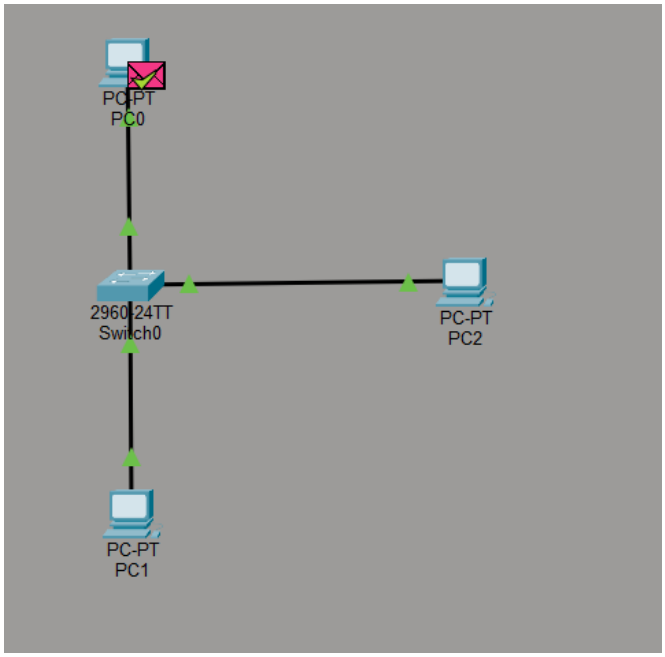
Observed Event Sequence:-



Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.004	Switch0	PC0	ARP
	0.004	--	PC0	ICMP
	0.005	PC0	Switch0	ICMP
	0.006	Switch0	PC1	ICMP
	0.007	PC1	Switch0	ICMP
	0.008	Switch0	PC0	ICMP
	1.008	--	PC0	ICMP
	1.009	PC0	Switch0	ICMP
	1.010	Switch0	PC1	ICMP
	1.011	PC1	Switch0	ICMP
	1.012	Switch0	PC0	ICMP
	2.013	--	PC0	ICMP
	2.014	PC0	Switch0	ICMP
	2.015	Switch0	PC1	ICMP
	2.016	PC1	Switch0	ICMP
	2.017	Switch0	PC0	ICMP
	3.018	--	PC0	ICMP
	3.019	PC0	Switch0	ICMP
	3.020	Switch0	PC1	ICMP
	3.021	PC1	Switch0	ICMP
	3.022	Switch0	PC0	ICMP





Task 3: Second Ping Observation

Without clearing the ARP cache, a second ping was issued immediately: ping 192.168.20.2. The simulation was re-run from the beginning of the event list for the new set of packets.

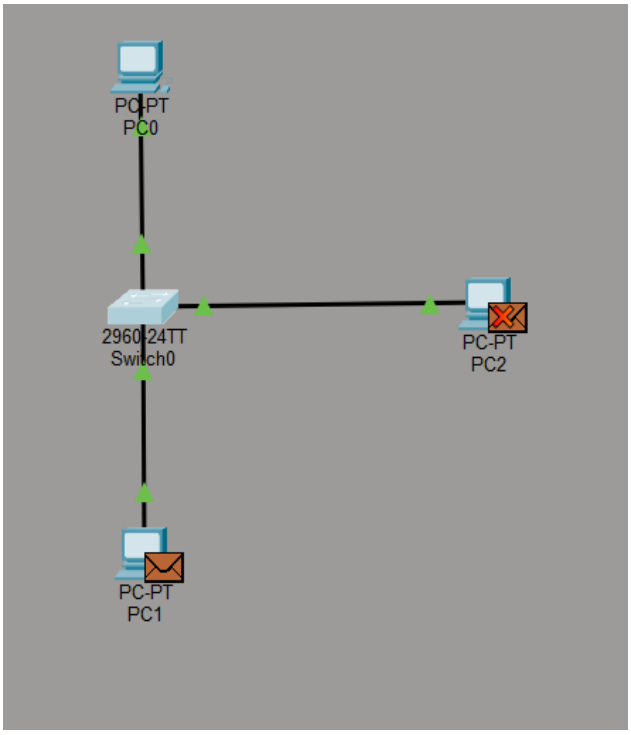
```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms
```

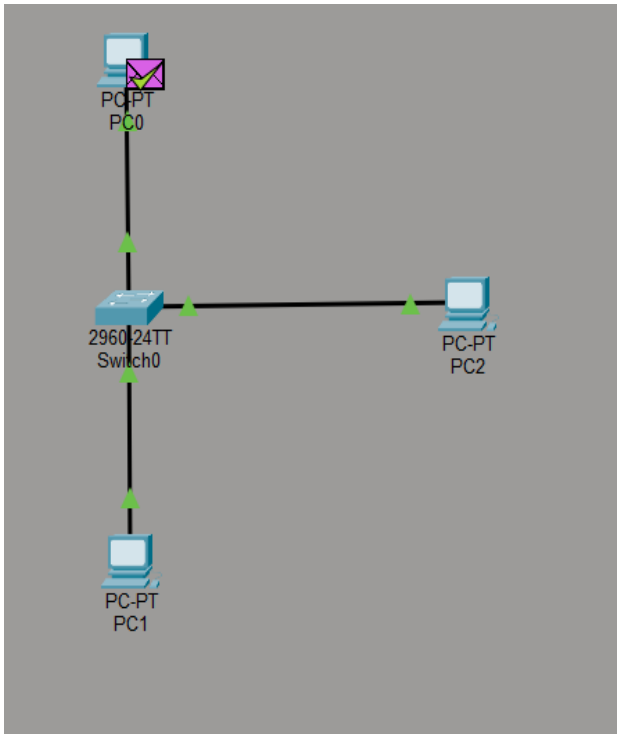
Observed Event Sequence:-

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.002	Switch0	PC1	ICMP
	0.002	Switch0	PC2	ICMP
	0.003	PC1	Switch0	ICMP
	0.004	Switch0	PC0	ICMP
	1.006	--	PC0	ICMP
	1.007	PC0	Switch0	ICMP
	1.008	Switch0	PC1	ICMP
	1.009	PC1	Switch0	ICMP
	1.010	Switch0	PC0	ICMP
	2.014	--	PC0	ICMP
	2.015	PC0	Switch0	ICMP
	2.016	Switch0	PC1	ICMP
	2.017	PC1	Switch0	ICMP
	2.018	Switch0	PC0	ICMP
	3.018	--	PC0	ICMP
	3.019	PC0	Switch0	ICMP
	3.020	Switch0	PC1	ICMP
	3.021	PC1	Switch0	ICMP
	3.022	Switch0	PC0	ICMP





Task 4: MAC Table Check

To verify switch learning, the MAC address table was inspected via the switch CLI in Packet Tracer

```
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.166e.0c77    DYNAMIC     Fa0/2
   1    0005.5e0c.9354    DYNAMIC     Fa0/1
```

- **First Ping vs. Second Ping – Comparison**

Feature	First Ping	Second Ping
ARP Involvement	Required	Not required
Response Time	Takes more time	Much quicker
Number of Packets	Higher	Lower
Explanation	MAC address needs to be discovered	MAC address already stored in cache